SMART SEASONAL FRUIT MARKETPLACE

Project Abstract

Smart Seasonal Fruit Marketplace is a full-stack e-commerce platform designed to help customers discover and purchase fresh seasonal fruits online. Unlike a conventional fruit-selling website, the platform dynamically highlights fruits according to the current season, availability, stock, and pricing. The system can promote products such as mangoes and watermelons during summer and automatically shift its recommendations as seasons change.

The platform provides customers with a modern shopping experience that includes product discovery, seasonal recommendations, search and filtering, product details, cart management, online ordering, order tracking, and personalized suggestions. An intelligent recommendation layer can prioritize fruits that are currently in season, locally available, or nearing the end of their optimal selling period. This helps customers find fresher products while helping sellers manage seasonal inventory more effectively.

An administrator dashboard allows authorized users to manage fruits, categories, seasonal availability, prices, stock, orders, users, and promotional offers. The system can also provide basic sales and inventory analytics to support better business decisions. The application is designed to be responsive so that customers can use it comfortably on mobile phones, tablets, and desktop computers.

The proposed technology stack uses Next.js for the web application and backend API layer, TypeScript for type-safe development, PostgreSQL for structured data, and a modern responsive UI. The application can be deployed as a publicly accessible website using services such as Vercel for the application and a managed PostgreSQL provider for the database. The result is intended to be a practical, scalable, and visually modern seasonal-commerce platform rather than a basic static website.

Key Features

- Season-based fruit discovery and recommendations
- Product search, filtering, and sorting
- Shopping cart and online ordering
- User accounts and order history
- Stock and price management
- Admin dashboard and sales analytics
- Responsive mobile-friendly design
- Deployable public web application